ROHAN MANCHANDA

Gurugram, Haryana, India | manchanda.rohan09@gmail.com | +91 99584 44171 | LinkedIn

EDUCATION

Heritage International Xperiential School, Gurugram *May 2025 – Present*

IB Diploma Programme — HL: Economics, Mathematics AA, Physics, Spanish Ab Initio; SL: Business Management, English

Grade 11 - Top 1%

Lancers International School, Gurugram *Jul 2019 – May 2025*

➔ IGCSE — Grade 10: 8 A*, 1 A | **Academic Excellence Award (Top 1% of 250+ students)**

Standardized Testing: SAT 1560 (Math 800 · EBRW 760)| APs: Physics C – Mechanics (5), Physics C – E&M (5), Calculus BC (5), Microeconomics (5)

RESEARCH & FELLOWSHIPS

Research Fellow — AI Summer Fellowship, MIT Media Lab *Jun – Jul 2025*

➔ Built CarePal, an AI medical assistant that collects patient symptoms, detects contradictions, and generates structured clinical reports, under the mentorship of MIT Prof. Ramesh Raskar.

➔ Currently leading efforts to pilot CarePal in hospitals across the Delhi NCR and Kolkata regions.

Research Investigator — National Centre for Coal and Energy Research *Jun 2024 – Jun 2025*

➔ Selected as the programme's **only high-school intern** across two consecutive summer fieldwork assignments.

➔ Investigated safety gaps and operational risks at 15+ collieries in the Jharia coal belt, a region affected by 68 active underground fires.

➔ Analysed ventilation failures, methane accumulation, fire-suppression systems, and emergency preparedness; authored an investigative **memorandum submitted to the Ministry of Coal.**

➔ Collaborated with geological-survey teams and disaster-response personnel during live rescue drills and emergency simulations.

Exchange Rate Dynamics & Recent Shocks in India

- Conducted an econometric study with Prof. Colin Constantine (Cambridge) on the impact of demonetization, COVID-19, and the global financial crisis on INR/USD exchange-rate dynamics.
- Published the research in the *Academic Journal of Health Sciences and Research* and presented the methodology at a virtual Cambridge economics research symposium.

Stock Market Analysis — S.T. Yau High School Science Award (Asia)

- Ran advanced statistical analyses linking investor trading activity to market volatility and liquidity, surfacing patterns in institutional behaviour that shape capital flows in emerging markets.
- DOI - Impact of Institutional Investors and Mutual Funds on BSE Volatility
- Honourable Mention, Pan-Asia S.T. Yau (Economics & Financial Modeling); presented at SBMIT 2025 with an **Honourable Mention in the Youngest Researcher category**.

TECHNICAL PROJECTS & INNOVATION

SPINDLE — Hybrid Legged-Wheeled Mine-Rescue Robot ***(Patent Holder)***

- Designed a hybrid rescue-and-inspection robot for collapsed, hazardous mine environments, integrating methane detection, oxygen sensing, LiDAR, thermal imaging, and structural-integrity assessment.
- Built the prototype at ~85% below the cost of comparable commercial rescue robots and field-tested it in an active Jharia colliery under disaster-response supervision.
- Secured an Indian patent (publication pending); presented at the Sigma Xi International Research Conference and currently under evaluation for field-deployment trials.
- Recognised with the BSA CREST **Gold Award** and the **Youth Changemaker Award** for mine-safety innovation.

LEADERSHIP, ENTREPRENEURSHIP & IMPACT

Business Strategy Intern — Harvard Ventures Technology Scholars Program *Jul – Aug 2025*

- Selected into a competitive SEAS-TECH and Harvard Undergraduate Ventures program pairing students with VC-backed startups, with seminars from founders, investors, and faculty (past speakers include several Forbes 30 Under 30 honorees).
- Placed at DocuBridge.ai to lead a go-to-market, competitor-landscape, and customer-discovery project for an AI Excel add-in that automates financial modeling for private-equity and growth-equity teams.

Coal-Worker Welfare & Safety Advocate — All India Coal and Mines Workers Federation *Sep 2024 – Present*

- Supported 120 pension claims, 53 workplace-injury compensation cases, and 29 widow-entitlement claims.
- Built a worker-led safety-reporting system across five collieries in the Jharia and Bokaro regions, documenting 500+ hazards, ~73% of which were resolved by site management.
- Conducted field interviews across 12 mines and submitted policy recommendations to the Ministry of Coal on pension delays, workplace safety, and protocol enforcement.

Founder — LokGharana *Oct 2023 – Present*

- Founded a non-profit reviving lesser-known and dying musical instruments, reaching a 35k+ audience online and in person.
- Enabled 52+ musicians across 6+ Indian states to stage 30+ events (₹280k / ~$3,140 in ticket sales) and partnered with government schools and orphanages to deliver free music education.

Optimization Intern — Infonet Technologies Pvt. Ltd., Gurugram *Jan – Mar 2024*

- Modeled real-world delivery constraints (time, demand, vehicle capacity) and built linear-optimization algorithms that improved delivery scheduling and routing efficiency.

Co-Founder — Bake to Educate (India) *Oct 2021 – Oct 2023*

- Co-founded a philanthropic venture funding education for underprivileged children, raising ₹80,000+ through large-scale bake sales while managing operations, pricing, and outreach.

Athletics — Basketball, Heritage International Xperiential School *Aug 2024 – Present*

- Founded the school's inaugural league (60+ participants), ran training clinics, and secured equipment sponsorships.

HONORS & AWARDS

CREST Gold Award (Robotics) — Top 1% globally	2025
Indian Patent Holder — SPINDLE Hybrid Legged-Wheeled Rescue Robot	2025
S.T. Yau High School Science Award (Asia) — Top 24 Finalist & Honourable Mention, Economics & Financial Modeling	2024
International Research Olympiad — Semifinalist, Top 100 of 45,000+ registrants	2025
Youth Changemaker Award — Mine-Safety Innovation	2025
Youngest Researcher Award — SBMIT Conference	2025
Australian Mathematics Competition — Distinction (Top 20% globally)	2025

SKILLS & COURSEWORK

Programming: Python, SQL, C, C++, JavaScript, HTML, CSS
AI & Development: LangChain, LangGraph, OpenAI APIs, Streamlit, Flutter, Git, GitHub
Data & Quantitative Tools: Jupyter, Advanced Excel, MATLAB, Wolfram Alpha, PuLP, Gurobi
Robotics & Simulation: Arduino IDE, PyBullet, Robot Operating System (ROS)
Quantitative Methods: Linear optimisation, OLS, ARDL, ECM, Johansen cointegration, Granger causality, ARIMA, VAR, time-series analysis, statistical hypothesis testing
Data Visualisation: Matplotlib, Seaborn, Plotly
Languages: English (fluent), Hindi (native), Spanish (conversational), German (advanced academic proficiency)
Coursework: Harvard CS50W: Web Programming with Python and JavaScript · IB Extended Essay in Economics — "Econometric Analysis of Coal-Industry Labour Productivity and the Impact of Safety Regulation in India, 2000–2025"